

Mariana Figueiredo

Bioengineer

 Mariana Figueiredo |  mdfigueiredop@gmail.com |  +351 914 665 196 |  Viseu, Portugal

SUMMARY

Results-driven Bioengineer with +2 years of cumulative experience in industrial and laboratory settings through curricular internships. Focused on the practical application of bioengineering to solve real-world biotechnology problems. Seeking a challenging professional position to contribute to pioneering industry projects.

AREAS OF EXPERTISE

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|------------------------|----------------------------|-------------------------------|
| ✓ Alternative Proteins | ✓ Bioactive Peptides | ✓ Design of Experiments (DoE) |
| ✓ Functional Foods | ✓ Interfacial Science | ✓ Enzymatic Bioprocessing |
| ✓ Data Analysis | ✓ Nutrient Bioavailability | ✓ Downstream Processing |

PROFESSIONAL EXPERIENCE

MSc Graduate Researcher

Feb 2026 - Sep 2026

Laboratory for Process Engineering, Environment, Biotechnology and Energy

Porto, Portugal

- Developed a **valorization pathway** for the alkaline wastewater of an agar production industry, converting discarded seaweed protein into high-value **ACE-inhibitory peptides**.
- Optimized hydrolysis via **Response Surface Methodology (RSM)**, reaching a **3-fold increase** in the total inhibitory capacity of the effluent.
- Assessed peptide **bioaccessibility** through simulated gastrointestinal digestion, with >50% of the ACE-inhibitory activity retained after digestion.
- Engineered an innovative **immobilized-protease biocatalyst** reaching an **80.6% immobilization yield** after RSM optimization.
- Characterized free and immobilized enzyme by **Michaelis-Menten kinetic modeling** (non-linear regression), retaining 40-55% of the free enzyme specific activity.

Research Intern

Sep 2025 - Jan 2026

Laboratory of Food Chemistry and Biochemistry KU Leuven

Leuven, Belgium

- Evaluated the **foaming properties** of **plant-based proteins** through standardized foaming assays.
- Characterized **interfacial properties** using tensiometry assays to correlate adsorption kinetics and interfacial rheology with foaming performance.
- Enhanced **sustainability profile** of sponge cakes by reducing egg reliance by 25% through plant-based protein substitution while maintaining batter and cake integrity.

Research Intern

Feb 2025 - Jun 2025

Laboratory for Process Engineering, Environment, Biotechnology and Energy

Porto, Portugal

- Extracted and enzymatically hydrolysed **macroalgae-based proteins** to yield bioactive peptides.
- Evaluated **anti-hypertensive** potential through ACE-inhibitory activity assays, achieving a IC_{50} of 52.4 $\mu\text{g/mL}$ and 56.6 $\mu\text{g/mL}$ for *Ulva* sp. and *Mastocarpus stellatus* peptides.
- Identified a total of 30 different potential anti-hypertensive peptides through an *in silico* screening and simulated their bioaccessability and bioavailability.

Engineer Intern <i>AgroGrIN Tech</i>	Feb 2024 - May 2024 Porto, Portugal
- Contributed to the optimization of production workflows, identifying potential bottlenecks and proposing alternative unit operations. - Conducted equipment research and feasibility studies for certain unit operations to assess their potential integration in the company's production system.	

PROJECTS

Industrial Production of Alpha-Mannosidase <i>Faculty of Engineering of University of Porto</i>	Fev 2025 - Jun 2025
- Developed a patent-based industrial design for alpha-mannosidase production and purification for pharmaceutical use. - Managed the project lifecycle from initial market analysis and process description to process flow modeling, technical equipment sizing for upstream and downstream operations and economic analysis.	

EDUCATION

Master's Degree in Bioengineering <i>University of Porto</i>	Sep 2024 - Oct 2026
Bachelor's Degree in Bioengineering <i>University of Porto</i> ECTS Grade: B	Sep 2021 - Jun 2024

VOLUNTEERING

Member of the Mentoring Department <i>Já T'Explico</i> Responsible for planning and providing weekly educational guidance to children in need.	Oct 2023 - Oct 2024
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PROFESSIONAL DEVELOPMENT

Working with Clients <i>British Council Online</i>	2024
Foundations of Project Management <i>Google (Coursera) Online</i>	2024
From Basic to Advanced - Complete Microsoft Excel Course <i>Udemy Online</i>	2023

ADDITIONAL INFORMATION

Technical Proficiencies: LaTeX | JMP Software | MATLAB | Microsoft Excel | Python

Languages: Portuguese | English